

Multi-Agent Item Generation and Expert-Free Critic Validation

(Methods & Results excerpt for the NCT paper — drop-in for the .docx)

How to use this file. Everything below maps 1:1 to the *Materials and Methods* and *Result and Discussion* sections. Grey boxes mark where to paste a figure/screenshot. All numbers and tables are the real results from our runs.

1 Materials and Methods

1.1 System overview

We frame item generation as a two-agent loop. A *Generator* agent G proposes a multiple-choice item; a *Critic* agent C scores it on several pedagogical dimensions and either accepts it or returns targeted feedback, in which case G regenerates (up to R retries). For mathematics, a deterministic *symbolic verifier* runs before the critic. The accepted item, its critic verdict, and (in ensemble mode) the per-critic breakdown are persisted as JSON and Markdown.

Fig. 1 — INSERT: agent workflow diagram

A block diagram: Generator $G \rightarrow$ (math only) Symbolic verifier $\rightarrow$ Critic / 3-model ensemble $\rightarrow$ decision $\langle \text{pass} \rangle$ store, or $\langle \text{fail} \rangle$ feedback loop back to G . Annotate the loop with “ $\leq R$ retries”.
Recommended: draw in draw.io / PowerPoint, export PNG.

1.2 Models and infrastructure

All agents are LLMs served through OpenRouter. We evaluate three frontier models as both generators and critics: GEMINI-3.1-PRO, GPT-5.5, and CLAUDE-SONNET-4.6. The generator samples at temperature 0.8 (diversity); the critic samples at 0.2 (stable scoring). Outputs are constrained to JSON.

1.3 Item specification and alignment to the national test

We align generation to the official National Testing Center (NTC, *Ulttyq testileu ortalygy*) teacher-certification specification. Each test form has 50 items: 40 standalone items and 10 context-based items organised as two shared-passage blocks of five questions each. Difficulty follows the official mix—basic (A), medium (B), hard (C)—in the proportions

$$p_A = 26\%, \quad p_B = 60\%, \quad p_C = 14\%.$$

For a batch of N items the per-level counts use largest-remainder rounding, $n_\ell = \text{round}(p_\ell N)$, $\sum_\ell n_\ell = N$. Topics follow the specification: 10 Kazakh-language topics (phonetics and orthography $\rightarrow$ lexicology $\rightarrow$ morphology $\rightarrow$ syntax $\rightarrow$ style/text analysis) and 20 mathematics topics.

1.4 Generator agent

The generator prompt is templated by subject, topic, and difficulty, and injects k few-shot exemplars retrieved from *real* published items of the same subject (and, for context blocks, anchored on a real context block). The model returns the stem, four options, the key, and an explanation.

Listing 1 / Fig. 2 — INSERT: example prompt + generated item

Paste one generator prompt (system message) and one resulting JSON item (e.g., a Kazakh lexicology item or a context block of five questions). *Use a monospaced screenshot or a shaded verbatim box in the .DOCX.*

1.5 Critic agent and the multi-critic ensemble

The critic first solves the item independently, then scores it on five dimensions $d \in \mathcal{D} = \{\text{correctness, distractor quality, difficulty alignment, language quality, LATEX/figure validity}\}$, each in $[0, 10]$. The overall score is a weighted mean

$$S = \frac{\sum_{d \in \mathcal{D}} w_d s_d}{\sum_{d \in \mathcal{D}} w_d}, \quad (w_{\text{correctness}} = 3, w_{\text{others}} \in \{1, 2\}), \quad (1)$$

and the item passes when $S \geq \theta$ (we use $\theta = 6.0$).

Ensemble (contribution 1). To reduce single-model self-evaluation bias we aggregate $K = 3$ critics. For the answer we take the majority vote $a^* = \text{mode}\{a^{(1)}, \dots, a^{(K)}\}$ with agreement $\alpha = \frac{1}{K} \sum_k \mathbf{1}[a^{(k)} = a^*]$; each dimension is the mean $\bar{s}_d = \frac{1}{K} \sum_k s_d^{(k)}$. Pairwise inter-critic reliability is reported with Cohen’s κ ,

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \quad (2)$$

where p_o is observed and p_e chance agreement on the chosen option.

1.6 Symbolic verification for mathematics (contribution 2)

For mathematics the generator may emit a machine-checkable specification that is evaluated in a sandboxed SYMPY process *before* the LLM critic. The verifier recomputes the answer symbolically; its verdict $v \in \{\text{confirmed, contradicted, inapplicable}\}$ is injected into the critic prompt as authoritative evidence, making correctness judgements model-independent for the items it covers.

1.7 Vision-aware critic

Figure-dependent (image) mathematics items are routed to a vision-capable critic that receives the rendered figure as a multimodal input; text items use the cheaper text critic. This ensures the critic can actually *see* the figure it is grading.

1.8 Expert-free critic validation: the Critic Discrimination Index (contribution 3)

We validate a critic *without* human grading by injecting known quality faults and measuring whether the critic detects them. From each real item we synthesise two degraded variants: WRONG-KEY (the key is relabelled to a distractor) and WEAK-DISTRACTOR (two distractors are replaced by implausible fillers). For dimension d the *Critic Discrimination Index* is the paired mean drop

$$\text{CDI}_d = \frac{1}{N} \sum_{i=1}^N s_d^{\text{real}}(i) - \frac{1}{N} \sum_{i=1}^N s_d^{\text{deg}}(i). \quad (3)$$

A well-calibrated critic yields a *large positive* CDI_d on the dimension the degradation targets (e.g. wrong-key $\rightarrow$ correctness) and $\text{CDI}_d \approx 0$ elsewhere (discriminant validity). Significance is assessed with the two-sided Wilcoxon signed-rank test, and effect size with the paired Cohen’s d ,

$$d = \frac{\bar{\Delta}}{s_{\Delta}}, \quad \Delta_i = s_d^{\text{real}}(i) - s_d^{\text{deg}}(i), \quad (4)$$

where $\bar{\Delta}$ and s_{Δ} are the mean and standard deviation of the paired differences ($d > 0.8$ is conventionally a large effect).

2 Result and Discussion

2.1 The critics discriminate quality (CDI)

Tables 1 and 2 report CDI for the targeted dimensions across the three critics. On both subjects and for all three models the discrimination is *statistically significant* with *large* effect sizes: relabelling the key collapses the correctness score, and weakening the distractors collapses the distractor-quality score, exactly as a valid critic should behave. Figure 1 visualises the gaps.

Table 1: CDI on **mathematics** ($N \approx 40$ real items per model; all $p < 0.001$, Wilcoxon).

Critic	Solve acc.	wrong-key $\rightarrow$ correctness gap (d)	weak-distr. $\rightarrow$ distractor q. gap (d)
Claude-Sonnet-4.6	0.98	+8.85 (4.94)	+3.03 (1.53)
Gemini-3.1-Pro	0.95	+8.58 (2.95)	+5.10 (1.50)
GPT-5.5	1.00	+8.73 (2.75)	+2.99 (1.69)

Table 2: CDI on **Kazakh language** ($N = 20$ real items per model; all $p \leq 0.0002$, Wilcoxon).

Critic	Solve acc.	wrong-key $\rightarrow$ correctness gap (d)	weak-distr. $\rightarrow$ distractor q. gap (d)
Claude-Sonnet-4.6	0.75	+5.15 (1.73)	+4.60 (3.87)
Gemini-3.1-Pro	0.90	+8.15 (1.87)	+7.90 (4.07)
GPT-5.5	0.80	+6.70 (1.69)	+3.90 (2.99)

2.2 Inter-critic agreement

On real items the three critics agree on the answer with high frequency; we report pairwise Cohen’s κ (Eq. 2) computed on the ensemble’s chosen options.

INSERT: Cohen’s κ values

Fill from the ensemble run’s `pairwise_kappa` (`summary.json`). One small table: critic-pair $\times$ κ .

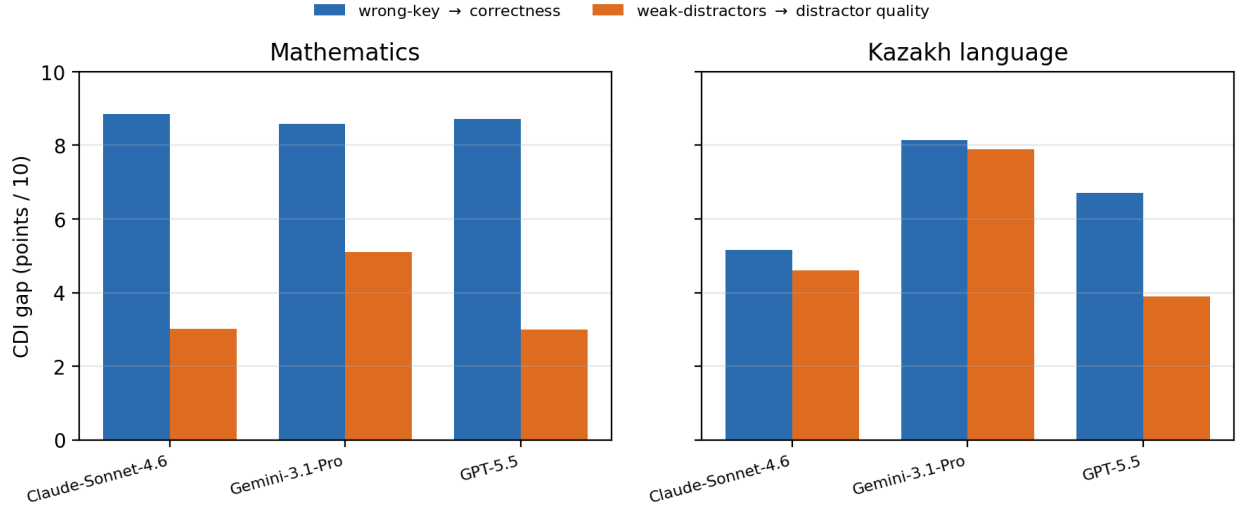


Figure 1: Critic Discrimination Index (gap between real and degraded mean scores) for the targeted dimensions, by model and subject. Larger is better; all gaps are statistically significant.

2.3 Vision matters for figure items (ablation)

On the figure-dependent mathematics items, critics evaluated *without* the image (text only) answered 6/9 correctly across the three models, whereas with the figure supplied to the vision critic they answered 9/9 (Table 3). This isolates the value of the vision-aware routing. The sample is small ($n = 3$ image items per model) and is reported as a controlled sanity check rather than a powered comparison.

Table 3: Vision ablation on figure-dependent mathematics items.

Critic	Blind (no image)	With image
Claude-Sonnet-4.6	2/3	3/3
Gemini-3.1-Pro	1/3	3/3
GPT-5.5	3/3	3/3
Total	6/9	9/9

2.4 Generated item bank

Using the validated ensemble critic we generated a full Kazakh-language bank of 150 items (three models $\times$ 50), Table 4. Standalone items passed the ensemble at 120/120 with a mean overall score of 9.14/10; for the six context blocks the ensemble’s independent answer matched the generated key on all 30/30 questions.

INSERT (optional): example generated context block

Paste one passage + its five questions (Kazakh) as an illustrative example, with the ensemble score and agreement per question.

Table 4: Generated Kazakh item bank (3 models $\times$ 50; ensemble critic).

Type	Items	Critic-key match	Passed
No-context (standalone)	120	—	120/120
With-context (6 blocks)	30	30/30	30/30
Total	150		

2.5 Discussion and limitations

The results support three claims: (i) a multi-agent generate-critique pipeline produces national-test-aligned items in a low-resource language at high accepted quality; (ii) CDI validates the critic *without* expert grading, showing significant, large-effect discrimination on both subjects; and (iii) an ensemble of heterogeneous critics plus symbolic verification reduces single-model bias. Limitations to state honestly: CDI is a *proxy* for human judgement (expert-correlation is future work); the vision ablation ($n = 3$) and the single real context block limit statistical power for those specific analyses; and the generated bank is validated by the automated critic rather than by human experts, which we plan to address with a targeted expert review.